

FUCINO, Italy

WMO: 162270

Lat: 41.9783N

Lon: 13.6053E

Elev: 651

StdP: 93.74

Time Zone: 1.00 (EUC)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.6	(c) -4.9	(d) -8.5	(e) 2.0	(f) -2.2	(g) -6.6	(h) 2.3	(i) -2.1	(j) 8.6	(k) 2.3	(l) 7.7	(m) 3.3	(n) 0.5	(o) 60	(p) 0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	32.6	17.2	30.9	17.1	29.4	17.2	22.5	26.5	21.7	26.5	20.8	26.2	2.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	17.3	24.6	20.2	16.1	24.3	19.0	14.9	23.9	69.7	26.4	66.6	26.5	63.4	26.3	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.3	(b) 6.2	(c) 5.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 7.3	(b) 6.2	(c) 5.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	3.5	5.0	7.1	11.2	15.6	18.9	21.3	21.0	16.8	13.1	7.6	3.8		
	DBStd	7.09	3.38	3.23	3.00	3.66	2.95	3.14	2.29	2.44	3.17	3.31	3.55	3.23		
	HDD10.0	766	204	142	97	30	1	0	0	0	1	11	88	193		
	HDD18.3	2531	461	374	349	214	92	30	4	6	67	163	321	450		
	CDD10.0	1537	2	1	6	66	175	266	350	341	203	108	17	2		
	CDD18.3	261	0	0	0	1	9	46	96	89	20	1	0	0		
	CDH23.3	3784	0	0	1	46	213	669	1314	1170	324	47	1	0		
	CDH26.7	1203	0	0	0	1	27	191	477	432	73	2	0	0		
(14) Wind	WSAvg	1.3	1.2	1.5	1.6	1.4	1.3	1.2	1.3	1.3	1.2	1.1	1.0	1.3		
(15) Precipitation	PrecAvg	889	80	68	78	80	57	36	30	31	73	108	135	114		
	PrecMax	1183	230	146	211	138	111	101	71	84	147	237	327	242		
	PrecMin	585	3	10	0	35	15	5	1	3	29	8	37	2		
	PrecStd	167	56	37	48	30	26	23	19	24	36	63	71	65		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	18.4	21.6	26.5	28.9	33.0	35.3	34.7	31.1	26.6	21.0	14.4		
		MCWB	11.3	11.3	13.5	15.9	17.1	18.8	19.2	17.7	16.4	17.8	13.6	10.2		
	2%	DB	13.3	15.5	19.4	24.8	27.3	30.6	32.7	32.5	29.0	24.6	18.5	12.7		
		MCWB	9.8	10.0	11.9	15.6	16.7	17.4	16.8	16.8	16.4	16.5	12.4	9.6		
	5%	DB	11.0	13.5	17.3	22.9	25.9	29.0	30.7	30.9	27.4	23.1	16.4	11.1		
		MCWB	8.4	9.3	10.9	14.8	16.7	17.3	17.1	17.0	16.8	16.1	11.3	8.7		
	10%	DB	9.1	11.5	14.6	20.2	24.2	27.2	29.2	29.2	25.5	21.0	14.2	9.3		
		MCWB	7.5	8.3	9.6	13.3	15.7	17.1	17.4	17.0	16.5	14.6	10.3	7.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	13.5	15.3	17.7	21.5	23.3	23.1	23.6	22.3	20.0	17.1	11.6		
		MCDB	15.8	18.1	21.3	23.5	25.0	27.3	26.1	26.3	25.2	24.0	20.0	13.2		
	2%	WB	11.3	11.4	13.4	16.7	19.9	22.1	22.2	22.3	21.0	18.8	14.0	10.6		
		MCDB	14.0	14.6	18.2	23.5	24.8	27.0	26.7	26.6	25.0	23.1	17.4	12.7		
	5%	WB	9.7	10.3	11.8	15.7	18.5	21.0	21.4	21.3	19.9	17.7	12.0	9.7		
		MCDB	12.2	14.0	16.6	22.1	24.2	26.2	26.8	26.8	25.1	22.1	15.5	11.7		
	10%	WB	8.7	9.3	10.5	14.5	17.1	19.7	20.5	20.1	18.2	16.4	10.9	8.9		
		MCDB	10.7	12.9	15.8	20.0	22.4	25.0	26.6	26.2	23.4	20.5	15.4	10.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	12.0	13.0	13.8	14.0	14.7	15.9	16.0	14.4	14.5	11.2	8.4		
		MCDBR	12.9	16.2	19.2	19.1	16.7	17.0	17.8	18.5	17.8	17.8	15.1	10.5		
		MCWBR	10.2	10.8	13.7	11.2	7.0	6.0	5.6	5.6	6.4	10.9	9.6	8.2		
	5% WB	MCDBR	12.7	13.8	18.0	17.0	14.6	14.6	15.1	15.3	14.9	16.1	14.0	8.6		
		MCWBR	10.2	10.3	12.8	10.5	9.0	8.4	7.8	8.5	9.2	11.3	10.5	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.343	0.384	0.425	0.431	0.428	0.418	0.417	0.403	0.376	0.350	0.319		
	taud		2.434	2.331	2.230	2.129	2.158	2.215	2.252	2.263	2.288	2.342	2.360	2.431		
	Ebn at Noon		827	854	855	843	847	849	853	842	826	804	771	788		
	Edh at Noon		78	100	125	149	148	140	134	129	117	98	83	72		
(44) All-Sky Solar Radiation	RadAvg		1.72	2.49	3.62	4.71	5.72	6.63	6.95	6.13	4.40	3.07	1.92	1.55		
	RadStd		0.20	0.28	0.46	0.39	0.53	0.37	0.42	0.43	0.25	0.31	0.21	0.20		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-79	+131	+68

Nomenclature: See separate page